NAME: _____ Geometry (FL B.E.S.T.)

5.3 Indirect Measurement

Similar triangles let us find heights and distances that are impractical to measure directly. In the **shadow method**, the sun's rays are essentially parallel, so an object and its shadow form a right triangle similar to any other object-and-shadow triangle measured at the same moment (AA: both right angles, plus the shared angle of elevation to the sun). In the **mirror method**, the angle of incidence equals the angle of reflection at a mirror on the ground, creating two similar right triangles. Surveyors use the same idea to measure distances across rivers or ponds using sightlines and stakes. In every case, the key step is setting up a correct proportion that matches height-to-height and base-to-base (never height-to-base). Some problems add an extra step, such as an observer's eye height, after the proportion is solved. This worksheet practises all of these indirect-measurement setups with full multi-step solutions.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. A 5 ft student casts a 3 ft shadow; a tree casts an 18 ft shadow. Find the tree height.

2. A 2 m stick casts a 1.5 m shadow; a pole casts a 9 m shadow. Find the pole height.

3. Mirror is 6 ft from you and 30 ft from a wall. Your eye height is 5 ft. Find the wall height.

4. $\frac{h}{45} = \frac{8}{12}$. Find h.

5. A 4 ft child casts a 5 ft shadow; a building casts a 40 ft shadow. Find the building height.

6. On a drawing, 1 in represents 20 ft. A room is 3.5 in long on the drawing. Find the real length.

7. A 12 m tower and a 3 m pole cast shadows at the same time. The pole's shadow is 2 m. Find the tower's shadow.

8. A ramp rises 2 m over a 10 m horizontal run. How high is it after 15 m horizontally?

9. Surveying proportion: $\frac{d}{75} = \frac{8}{10}$. Find d.

10. Mirror 10 ft from a person (5.5 ft eye height) and 40 ft from a statue. Find the statue height.

11. Person 6 ft tall casts an 8 ft shadow; a tree's shadow is 28 ft. Find the tree height.

12. 3 cm on a plan represents 12 m. A wall measures 5 cm on the plan. Find the real length.

13. A student's eye height is 5 ft, standing 40 ft from a tower. The line of sight forms a right triangle similar to a reference triangle with legs 3 (rise) and 4 (run). Find the total tower height.

14. A pole of height $(2x + 1)$ m casts a 6 m shadow at the same time a 10 m tower casts a 15 m shadow. Find x and the pole's height.

15. To find pond width w, similar triangles give $\frac{w}{18} = \frac{24}{16}$. Find w, then find the total distance to a dock that is 5 m past the far shore.

✓ Answer Key

1. 30 ft
2. 12 m
3. 25 ft
4. 30
5. 32 ft
6. 70 ft
7. 8 m
8. 3 m
9. 60
10. 22 ft
11. 21 ft
12. 20 m
13. 35 ft
14. $x = 1.5$; pole height = 4 m
15. $w = 27$ m; total distance = 32 m